
Digital Egypt Pioneers Initiative - Data Analysis Track



GRADUATION PROJECT DOCUMENTATION

Bitcoin Analysis project
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1

Project Overview

1 Project Overview

The Bitcoin cryptocurrency market represents one of the most dynamic and data-rich financial environments in the world. Since its inception in 2009, Bitcoin has grown from a niche digital asset into a globally traded financial instrument, generating continuous streams of high-frequency market data across multiple timeframes. Understanding the behavioral patterns, volatility cycles, and trading dynamics embedded within this data has become increasingly valuable for researchers, analysts, and market participants alike.

1.1 Project Concept

This project — **Bitcoin Analysis** — is a data analysis project developed as part of the Digital Egypt Pioneers Initiative (DEPI), Data Analysis Track. The project leverages a comprehensive historical Bitcoin OHLCV dataset sourced from Kaggle, spanning from 2018 to 2026, and covering four distinct trading timeframes: 15 minutes, 1 hour, 4 hours, and 1 day. This multi-timeframe structure allows for analysis at both the micro and macro levels of market behavior.

The analysis pipeline was built in Python, following the **Medallion Architecture (Bronze → Silver → Gold)**, then we built a Machine Learning model to predict whether the next Bitcoin price will increase or decrease based on historical market data. and culminates in an interactive **Power BI dashboard** designed to surface actionable market insights across four analytical domains.

The motivation behind this project is twofold:

- **Analytical:** To apply the full data analysis pipeline from raw ingestion and cleaning to dimensional modeling, and interactive visualization to a real world, financially significant domain, while rigorously avoiding a critical technical pitfall: the cross-timeframe contamination that arises when candles of different durations are mixed in rolling calculations without proper grouping.
- **Market Intelligence:** To produce insights that identify recurring patterns in Bitcoin's price behavior, volatility cycles, whale activity, and market consolidation phases information that can support informed decision-making and a deeper understanding of cryptocurrency market dynamics.

Project objectives span five domains:

- **Data Engineering:** Ingest four raw Bitcoin OHLCV CSVs into the Bronze layer, then clean, validate, and deduplicate them in the Silver layer.
- **Feature Engineering:** Derive a comprehensive set of market KPIs including price movement, market volatility, trend direction and persistence, buy/sell dominance, order flow imbalance, whale activity detection, breakout signals, support and resistance distances, market overheating and compression phases
- **Dimensional Modeling:** Design and populate a Star Schema data model in the Gold layer, consisting of a central fact table (fact_bitcoin_market) and supporting dimension tables (dim_date, dim_timeframe).
- **Dashboard Development:** Build a four-page interactive Power BI dashboard covering Market Overview, Buyer vs Seller Pressure, Liquidity, Support & Resistance, Trend & Breakout Tracking, Trader Activity & Whale Watch
- **Insight Generation:** Identify statistically meaningful patterns in Bitcoin's market behavior across eight years of data (2018–2026)

1.2 Data Source

The dataset used in this project was downloaded from **Kaggle** and contains historical Bitcoin trading data from **2018 to 2026**.

Attribute	Detail
Dataset Name	BITCOIN Historical Datasets 2018-2026 Binance API
Source Platform	Kaggle
Dataset URL	https://www.kaggle.com/datasets/novandraanugrah/bitcoin-historical-datasets-2018-2024
Files count	4 files with four different timeframes 15-minute intervals (15m) 1-hour intervals (1h) 4-hour intervals (4h) 1-day intervals (1d)
Total Rows Available at the time we used the data	391510 records (4 files companied)
Rows Used in Project	391510 records (full downloaded subset)
Files Format	CSV

Each file has 12 columns:

Column Name	Type	Description
Open time	Object	Opening timestamp in UTC
Open	float64	Opening price
High	float64	Highest price during period
Low	float64	Lowest price during period
Close	float64	Closing price
Volume	float64	Trading volume
Close time	Object	Closing timestamp in UTC
Quote asset volume	float64	Volume in quote asset
Number of trades	int64	Number of trades during period
Taker buy base asset volume	float64	Volume of taker buy orders

Taker buy quote asset volume	float64	Volume of taker buy orders in quote asset
Ignore	int64	Unused field

1.3 Project Impact

This project supports data-driven cryptocurrency market analysis by transforming large volumes of historical Bitcoin data into clear analytical insights and predictive signals. The generated dashboards and machine learning model simplify market monitoring, helping analysts and traders evaluate market conditions and recognize potential opportunities more efficiently.

1.4 Main Charts Overview

The four-page Power BI dashboard is structured into four analytical sections, each focusing on a specific aspect of Bitcoin market behavior and performance.

Page	Title	Focus Area	Key KPIs Shown
1	Market Overview	Overall price behavior and market activity at a glance	Price Movement, Volatility, Volume Spikes, Capital Flow
2	Buyer vs Seller Pressure	Balance of buying and selling activity driving the market	Buy VS Sell Pressure, Net Order Flow, Overheating Signal
3	Liquidity, Support & Resistance	Where trading volume concentrates and how price behaves around key levels	Key Level Volume, Volume by Price Level, Avg Price Rejection, Price Rejection, Avg Support Strength, Support Strength, Consolidation Range
4	Trend & Breakout Tracking	Strength and durability of price trends, and potential breakout points	Trend Strength, Trend Persistence, Breakout Status, Compression
5	Trader Activity & Whale Watch	Trading participation patterns and detection of large/concentrated trades	Trade Count, Trade Participation, Trade Intensity, Whale Flags, Volatility Clusters
6	The Forecast	Forward-looking projection of Bitcoin price direction	Prediction, Confidence, Accuracy, Precision/Recall, Feature Importance, Prediction Confidence, Confusion Matrix, Prediction by Time Frame

1.5 SMART Questions

The analysis is guided by five SMART (Specific, Measurable, Achievable, Relevant, and Time-bound) research questions derived from the project objectives and the Bitcoin market dataset.

#	SMART Question
1	How did Bitcoin market performance (price movement, volatility, capital flow, and volume spikes) change across the selected time period?
2	How did buyer and seller dominance change during the selected time period, and when was market overheating most significant?
3	How did market liquidity, support strength, resistance strength, and market consolidation vary during the selected time period?
4	How did trend persistence, trend strength, and breakout events evolve during the selected time period?
5	How did trade activity, trade intensity, whale activity, and sustained high-volatility periods change during the selected time period?

2

Technical Architecture

2 Technical Architecture

This chapter documents the tools, architectural patterns, and data model that form the technical foundation of the project. The data pipeline was developed using Python following the Medallion Architecture, and its output is a star schema designed for Power BI analytics and reporting.

2.1 Tools Used

Several tools were used throughout the project ,from data exploration, preprocessing, and feature engineering, through dimensional modeling, to machine learning and dashboard development:

Tool / Platform	Purpose	Project usage
Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn)	Core programming language	Used for exploratory data analysis (EDA), data cleaning, feature engineering, implementing the Medallion Architecture (Bronze, Silver, and Gold), building the star schema, and developing and evaluating the machine learning model.
Power BI	Business intelligence and data visualization	Used to build the interactive dashboard, create KPI cards and visualizations, and present analytical insights from the Gold layer.
Figma	UI/UX design & dashboard wireframing	Used to design the dashboard layout, visual hierarchy, and page structure before implementing the final dashboard in Power BI.
GitHub	Project repository	Used to publish and share the project source code, documentation, and related project files.

2.2 Architecture (Medallion)

The Medallion Architecture organizes data into three logical layers Bronze, Silver, and Gold each serving a specific purpose. This architecture ensures that raw data is preserved in the **Bronze** layer, while the **Silver** layer centralizes data cleaning, timeframe merging, and feature engineering. The **Gold** layer provides a structured star schema optimized for reporting, dashboard development, and machine learning.

<u>Bronze Layer</u> Raw data ingestion No transformation	<u>Silver Layer</u> Timeframe merging, Data cleaning & Feature engineering	<u>Gold Layer</u> Star schema 2dim + 1 fact
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2.3 Data Model (Star Schema)

The project uses a **Star Schema** to organize the data for reporting and analysis.

The data model consists of:

Table	Type	Key Columns	Description
fact_bitcoin_market	Fact	date_key, timeframe_key	Contains all market data and engineered features
dim_date	Dimension	date_key	Contains date information such as year, quarter, month, and day.
dim_timeframe	Dimension	timeframe_key	Contains the available trading timeframes

These relationships allow Power BI to filter and analyze the data efficiently.

Chapter 3

Data Engineering

3 Data Engineering

This chapter documents the complete data preparation and transformation pipeline across the Medallion Architecture layers, covering data ingestion, cleaning, and transformation processes used to prepare the final analytical dataset.

3.1 Ingestion & Cleaning Process

The cleaning process spans the Bronze and Silver layers. Bronze ingests and preserves raw data; Silver applies all corrections and validations.

Bronze Layer – Raw data Ingestion

In the Bronze layer, the raw Bitcoin datasets were downloaded from Kaggle using the KaggleHub library. Multiple time-based datasets (15-minute, 1-hour, 4-hour and 1-day intervals) were loaded as separate CSV files using Pandas. Each dataset was saved as a raw copy to preserve its original structure for future processing. Basic exploratory checks such as `shape()`, `head()`, and `info()` were performed to understand the dataset structure and validate its quality. No transformations were applied at this stage.

Link: [Bronze layer](#)

Silver Layer - Data Cleaning

The Bronze datasets from all supported timeframes (15 Minutes, 1 Hour, 4 Hours, and 1 Day) were combined into a single dataset after assigning a timeframe identifier to each record. Date columns were converted into datetime format using UTC, and the dataset was sorted chronologically to preserve the correct sequence of market events.

attributes including Year, Quarter, Month, and Day were extracted from the Open Time column to simplify time-based analysis within Power BI.

The dataset was then validated by examining its structure, data types, duplicate records, and invalid values. Duplicate observations were identified using the combination of Open Time and Timeframe,

3.2 Feature Engineering

Silver Layer

After completing the data cleaning process, a comprehensive set of analytical features was generated to enrich the dataset and better represent Bitcoin market behavior. These engineered features transform the original trading data into meaningful business indicators that support advanced analysis, dimensional modeling, and Power BI dashboard development.

- **Price KPIs**

A collection of price-related indicators was created to measure market performance and price movement during each trading interval. These features capture changes in price, percentage returns, and market volatility.

Generated columns:

- **price_change**
- **return_pct**
- **volatility**

- **Volume KPIs**

Trading volume was further analyzed by generating additional indicators that describe buying activity, selling activity, capital flow, and order flow. These features provide deeper insights into market participation and trading pressure.

Generated columns:

- **capital_flow**
- **buy_volume**
- **sell_volume**
- **buy_ratio**
- **sell_ratio**
- **order_flow**

- **Trend Indicators**

Trend-related features were generated to identify market direction, measure how long a trend persists, and evaluate the strength of each trend.

Generated columns:

- **trend_direction**
- **trend_length**
- **trend_strength**

- **Support & Resistance**

Technical indicators were created to estimate support and resistance levels.

Generated columns:

- **support**
- **resistance**

- **Breakout Signals**

Breakout indicators were generated by comparing the closing price with rolling highs and lows.

Generated columns:

- **rolling_high_20**
- **rolling_low_20**
- **breakout_up**
- **breakout_down**

- **Market Behavior Signals**

Additional behavioral indicators were engineered to detect important market conditions and abnormal trading activity.

Generated columns:

- **avg_volume_20**
- **avg_volatility_20**
- **avg_trade_activity_20**
- **high_volatility**
- **high_volatility_length**
- **compression**
- **volume_spike**
- **overheating**
- **whale_threshold**
- **whale_activity**

Before exporting the Silver Layer, a final validation step was performed to ensure the completeness and reliability of the engineered features.

Rows containing null values generated by rolling-window calculations were removed. These null values occurred because rolling metrics require a sufficient number of previous observations before they can be calculated. Therefore, the earliest records in each timeframe did not have enough historical data to produce valid rolling indicators.

After removing these records, only the analytical columns required for downstream processing were retained.

As a result, four cleaned and feature-engineered dataset were generated. This dataset provide a standardized and analytics-ready foundation for the Gold Layer .

Link: [Silver Layer](#)

Gold Layer - Star Schema Build

The Gold layer contains the final analytical dataset organized as a Data Warehouse. The data is stored in a Star Schema, making it easy to create dashboards and train machine learning models.

link: [Gold Layer](#)

4

Data Analysis

4 Data Analysis

The Data Analysis stage was conducted to explore Bitcoin market behavior, identify meaningful patterns, and generate business insights before developing the interactive Power BI dashboards. This phase combines exploratory data analysis (EDA) with visualization techniques to better understand price movement, trading activity, market volatility across different timeframes.

The visualizations and KPIs developed in the business analysis serve as the analytical foundation for the interactive Power BI dashboards implemented in the final phase of the project.

Data Preparation

The analysis began by importing the required Python libraries for data manipulation and visualization, including Pandas, NumPy, Matplotlib, and Seaborn.

The Gold Layer was loaded using the Star Schema model, which consists of the Bitcoin Market Fact table along with the Date and Timeframe dimensions.

The fact table was then merged with both dimensions using `date_key` and `timeframe_key`. Row-count validation was performed after each merge to ensure that no duplicate records were introduced during the integration process.

Finally, the merged dataset was inspected by reviewing its structure, descriptive statistics, and data types before beginning the exploratory analysis.

4.1 Exploratory Data Analysis (EDA)

The purpose of the Exploratory Data Analysis stage was to better understand Bitcoin market behavior before designing the dashboards..

Chart name	Purpose	Insight
Price Change & Volatility Distribution	Analyze the distribution of price change and volatility across different timeframes to identify market variability and detect unusual observations.	Price change and volatility vary significantly across the historical dataset, with extreme values representing major Bitcoin market events rather than data quality issues.
Trend Length Distribution	Understand how long market trends usually persist before changing direction.	Most market trends remain short-lived, with the majority lasting only one or two consecutive candles. Longer trends occur much less frequently across all timeframes.
Buy Ratio Distribution	Evaluate buyer dominance and overall market balance.	The Buy Ratio remains concentrated around 50%, indicating that buyers and sellers are generally balanced. Strong buyer or seller dominance

		occurs only during specific market events.
Trade Activity Distribution	Compare trading activity across different timeframes and identify exceptional trading periods.	An extreme outlier appears within the daily timeframe, indicating an unusually active trading day that may correspond to a significant market event.
Average Volatility Over Time	Analyze how market volatility changes throughout different market cycles	Bitcoin volatility follows a cyclical pattern, with major peaks during highly active market periods and relatively stable behavior during quieter years.
Correlation Analysis	Investigate the relationships between the main market variables	Trading volume, capital flow, and trade activity exhibit strong positive relationships, while short-term price movement shows relatively weak linear correlation with the remaining variables.

4.2 Business Analysis (Dashboards)

The following section summarizes the business analyses performed in Python using the engineered KPIs. These analyses answer the key business questions and were later implemented as interactive Power BI dashboards.

1. Market Overview

Chart name	Business Question Answered	Definition	Logic
Price Movement	What is the overall price movement?	This chart shows the overall price movement during each time interval by measuring the difference between the Open and Close prices. It helps identify whether the market is in a bullish or bearish state over time and highlights short-term directional trends.	Displays the price_change values over time.
Volatility	How volatile is the market within each interval?	This chart measures market volatility by calculating the difference between the High and Low prices in each interval. Higher values indicate stronger price fluctuations and increased market risk, while lower values suggest more stable market conditions.	Displays the volatility values over time.
Capital Flow	How much capital is flowing through the market?	This chart shows the total capital flowing through the Bitcoin market over time. Higher values indicate increased market liquidity and participation, while lower values suggest reduced trading activity.	Calculated by summing the capital_flow values over the selected time period.
Volume Spikes	How many volume spike events occurred each period?	This chart shows the number of volume spike events detected each period. A volume spike occurs when trading volume becomes significantly higher than its recent average while the price also moves noticeably. These events often indicate increased market participation and may precede strong price movements.	Counts the total number of volume_spike events for each period.

2 . Buyer vs Seller Pressure

Chart name	Business Question Answered	Definition	Logic
Buyer vs Seller Pressure	Who dominated the market, buyers or sellers?	This chart compares the total Buy Ratio and Sell Ratio during the selected period, providing a snapshot of whether buyers or sellers dominated market activity.	Calculated by summing the buy_ratio and sell_ratio values over the selected period.
Net Order Flow	How did the dominance differ across periods?	This chart measures the net order flow by comparing buying and selling activity. Positive values indicate buyer dominance, while negative values highlight seller dominance.	Calculated by summing the order_flow values for each period.
Market Overheating	When did excessive buying activity peak?	This chart tracks the total number of overheating events detected each period. It helps identify the specific years where excessive buying activity was most frequent.	Counts the total number of overheating events for each period.

3 . Liquidity, Support & Resistance

Chart name	Business Question Answered	Definition	Logic
Resistance Strength	How did price rejection change across years?	This chart compares the average price resistance across years. Higher values indicate that prices were rejected more strongly from session highs before closing.	Calculated by averaging the resistance values for each period.
Support Strength	How did support strength change across years?	This chart compares the average support strength across years. Higher values indicate that buyers pushed prices farther away from session lows before the close.	Calculated by averaging the support values for each period.
Market Consolidation	Is the market consolidating?	This chart shows the annual percentage of time the market was in a compression state. Higher values indicate longer periods of market consolidation and sideways movement.	Calculated by averaging the compression values for each year and converting them into percentages.
Volume by Price Bucket	Where is the market most liquid?	This chart shows the total trading volume across different price ranges to identify where market liquidity was highest. Higher trading volume indicates greater market liquidity at that price level.	Calculated by grouping prices into fixed price buckets and summing the Volume within each bucket.

4 . Trend & Breakout Tracking

Chart name	Business Question Answered	Definition	Logic
Trend Strength vs Volume	Do increases in trading volume align with increases in trend strength?	This chart compares overall trading volume with trend strength to determine whether both metrics move together or diverge over time.	Calculated by summing both Volume and trend_strength for each year and displaying them together.
Trend Persistence	What was the maximum duration a market trend sustained each year?	This chart measures the maximum trend persistence by showing the longest continuous price move recorded each year. Higher values indicate years with highly sustained trends, while lower values suggest more frequent market reversals.	Calculated by selecting the maximum trend_length value for each year.
Breakout Status	Are breakout trends occurring?	This chart tracks the price levels where major market breakout trends occurred over time. It highlights the closing price of the latest breakout event detected each year, helping identify structural shifts in the Bitcoin market.	Calculated by selecting the latest breakout event (breakout_up or breakout_down) for each year and plotting its closing price.

5 . Trader Activity & Whale Watch

Chart name	Business Question Answered	Definition	Logic
Trade Participation	How active are traders in the market?	This chart shows the total trading activity during each period, helping identify whether market participation is increasing or decreasing over time. Higher values indicate greater market participation and stronger trading engagement.	Calculated by summing the trade_activity values for each year.
Trade Intensity	Is market volume driven by many small trades or a few large trades?	This chart compares the number of trades relative to trading volume, helping determine whether market activity is driven by many small trades or fewer large trades.	Calculated by dividing the total trade_activity by the total Volume for each year.
Whale Activity	Which periods experienced the highest whale activity?	This chart shows the total whale activity across each period to identify when large market participants were most active. Higher values indicate more intervals dominated by unusually large trading volume.	Calculated by summing the whale_activity flags for each year.
Volatility Clusters	Which periods experienced the longest sustained high-volatility phases?	This chart compares the longest continuous high-volatility periods, helping identify when market instability persisted for the longest duration.	Calculated by selecting the maximum high_volatility_length value for each year.

Note: The following analyses provide additional business insights and were not implemented in the final Power BI dashboards.

Chart name	Business Question Answered	Definition	Logic
Trend Persistence Distribution	How frequently do market trends persist across different lengths?	This chart shows the distribution of market trend lengths by counting how many times each trend duration occurred. Higher frequencies indicate that short-lived trends are more common, while lower	Calculated by counting the frequency of each trend_length value.

		frequencies represent rare but sustained market trends.	
Market Breakout Events	Which breakout trends dominated the market each year?	This chart compares the number of bullish and bearish breakout events detected each year. It helps identify whether upward or downward breakouts were more dominant during different market periods.	Calculated by summing the breakout_up and breakout_down events for each year and comparing their totals.

Link: [Data analysis](#)

5

Machine Learning

5 Machine Learning

5.1 ML Models Overview

The purpose of the machine learning model is to predict whether the Bitcoin price will experience a significant upward movement within the next few candles.

The model was built using the **Random Forest Classifier**, a supervised machine learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

To preserve the chronological order of financial data, the dataset was divided using a **time-based split** instead of a random split. Data from **2018–2023** was used for training, while data from **2024–2025** was used for testing. This approach simulates real market conditions, where future data is never available during training.

The prediction target was created by checking whether the maximum price within the next four candles exceeded a predefined return threshold. Different thresholds were used for each timeframe according to its volatility.

5.2 Model Results & Comparison

The model was trained separately for each trading timeframe because each timeframe has different market behavior.

The following table summarizes the model performance.

Timeframe	Training Period	Testing Period	Model
15 Minutes	2018–2023	2024–2025	Random Forest
1 Hour	2018–2023	2024–2025	Random Forest
4 Hours	2018–2023	2024–2025	Random Forest
1 Day	2018–2023	2024–2025	Random Forest

For each timeframe, the model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

These metrics provide a comprehensive evaluation of the prediction performance instead of relying only on accuracy.

5.3 Why Random Forest was Selected as the Prediction Model

Random Forest was selected because it performs well on structured tabular data and can capture complex relationships between market variables without requiring extensive parameter tuning.

The model also provides feature importance scores, making it easier to understand which financial indicators contribute the most to the prediction.

In addition, Random Forest is less sensitive to overfitting than a single decision tree because it combines the predictions of many trees.

Its advantages include:

- Easy to train and implement.
- Handles both numerical and binary features.
- Works well with engineered financial indicators.
- Provides feature importance rankings.
- Produces stable prediction results.

These characteristics make Random Forest a suitable baseline model for Bitcoin market prediction.

5.4 Forecast Model

After training the model, it was used to generate a prediction for the most recent market data.

The latest available market features are passed to the trained Random Forest model to predict whether a strong upward movement is expected within the next four candles.

The model produces:

- A prediction label:



- A confidence score representing the probability of a future price increase.

If the predicted probability exceeds the model threshold, the system generates a BUY Signal, indicating that the market is expected to experience a significant upward movement. Otherwise, the system returns NO BUY Signal, suggesting that the market is likely to remain bearish or move sideways.

This forecasting step demonstrates how the trained model can support traders by providing real-time predictions based on the latest market conditions.

Link: [ML](#)

6

Dashboard & Visualization

6 Dashboard & Visualization

The Dashboard & Visualization chapter presents the final end-user deliverable of the Bitcoin Analysis Project — a six-page interactive Power BI dashboard. This chapter covers the design philosophy behind the dashboard, the Figma prototyping process, the complete Power BI implementation, and a detailed walkthrough of each dashboard page.

6.1 Power BI Dashboard Pages

The Power BI dashboard 'Gaming & Mental Health' connects directly to the Gold layer Delta tables via the native Microsoft Fabric–Power BI integration. It is organized across six pages: a cover/intro page, an executive overview, and four thematic analytical pages. All calculations are driven by DAX measures defined in the data model.

The dashboard follows a deliberate data storytelling structure — beginning with a mission statement and context-setting, then progressing through executive-level KPIs before drilling into behavioral, psychological, physical, and productivity dimensions. The final page presents the machine learning prediction results as an interactive analytical page.

Page	Title
Cover	Bitcoin Analysis
1	Market Overview
2	Buyer VS Seller Pressure
3	Liquidity, Support & Resistance
4	Trend & Breakout Tracking
5	Trader Activity & Whale Watch
6	The Forecast

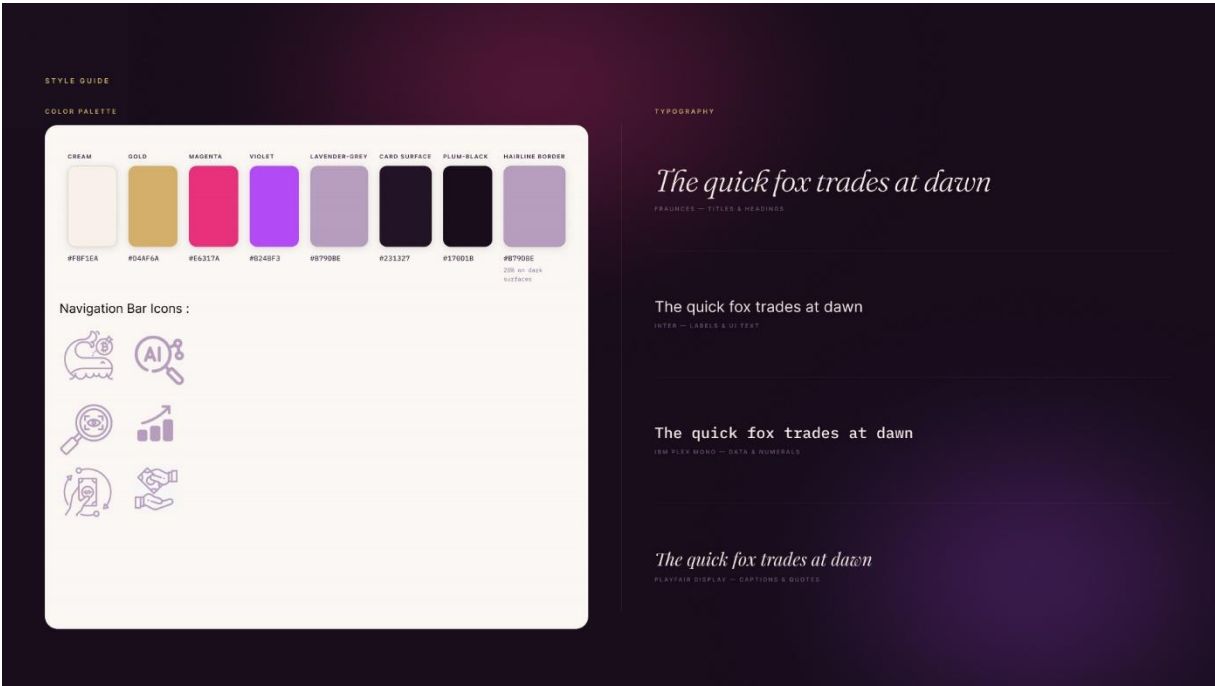
KPI	Value Extracted from 15m timeframe	Dashboard Page
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Avg. Volume	655.29	Market Overview
Avg. Capital Flow	19.05M	Market Overview
Return_Pct	3.47%	Market Overview
Avg. Volatility	153.82	Market Overview
Buy Pressure	96.27 M	Buyer VS Seller Pressure
Sell Pressure	97.62 M	Buyer VS Seller Pressure
Overheating Signal	5301	Buyer VS Seller Pressure
Key Level Volume	16.50K	Liquidity, Support & Resistance
Avg. Price Rejection	75.53	Liquidity, Support & Resistance
Avg. Support Strength	78.29	Liquidity, Support & Resistance
Consolidation Range	0.09	Liquidity, Support & Resistance
Trend Strength	-0.09	Trend & Breakout Tracking
Trend Persistence	16	Trend & Breakout Tracking
Breakout Status	66.16K	Trend & Breakout Tracking
Compression	0.09	Trend & Breakout Tracking
Total Trades	6 bn	Trader Activity & Whale Watch
Whale Flags	3006	Trader Activity & Whale Watch
Trade Intensity	33.05	Trader Activity & Whale Watch
Volatility Clusters	34	Trade Activity & Whale Watch
Prediction	0	The Forecast
Confidence	44.70%	The Forecast
Accuracy	62.39%	The Forecast
Precision/Recall	0.41/0.73	The Forecast

6.3 Figma Design

The dashboard's visual design was developed in Figma to establish a cohesive design system prior to implementation in Power BI. The interface follows a dark, editorial aesthetic that departs from conventional crypto/fintech dashboard styling (typically dark backgrounds paired with neon-green accents), instead using a warm, jewel-toned palette to give the project a distinct visual identity.

Figma Design



6.3.1 Visual Identity & Gaming Theme

Purpose	Color	Hex
Background	Deep plum-black	#170D1B
Panel/card background	Dark aubergine	#1C1121 / #12182B
Primary accent	Magenta	#E6317A
Secondary accent	Champagne gold	#D4AF6A
Tertiary accent	Electric violet	#B24BF3
Positive/bullish data	Mint green	#6EE7B7
Primary text	Cream white	#F8F1EA
Secondary/muted text	Lavender-grey	#B79DBE

Gold is used for numerical data and chart accents, magenta and violet are reserved for emphasis and negative/highlighted data points, and mint is used specifically to indicate positive or bullish values, keeping the color coding consistent across all pages.

Background Design

Each page uses a flat dark base color layered with two soft, blurred radial gradients (glow effects) — one in magenta positioned toward the top of the frame, and one in violet positioned toward the bottom — to create visual depth without introducing additional imagery or texture-heavy elements. A subtle grain/noise overlay is applied at low opacity across the full background to avoid a flat, overly digital appearance.

Typography

- **Headings/Titles:** Fraunces (serif) — used to give the interface an editorial, less generic feel compared to typical sans-serif dashboard typography.
- **Body and Labels:** Inter — used for supporting text and UI labels.
- **Data and Numerical Values:** IBM Plex Mono — used consistently across all KPI values and chart figures to reinforce a data-driven, analytical tone.

Following design revisions, KPI card titles and chart titles were unified to use the same font and font size across all pages, improving visual consistency and reducing hierarchy inconsistencies present in earlier drafts.

Navigation

The left-hand navigation sidebar was simplified during the design revision process. Icons were removed from all navigation items, leaving clean text-only labels for each page. This change was made to reduce visual clutter and maintain focus on the data visualizations rather than decorative elements.

Content Revisions

Several changes were made to the initial design concept to align it more closely with the project's analytical purpose:

- The working project title ("Lumen") was removed and replaced with **"Bitcoin Analysis Project"** as the primary title.
- Page-level decorative titles (e.g., stylized names such as "Inner Circle") were removed. Functional subtitles (e.g., **"Market Overview"**) were retained and repositioned to serve as the primary page titles, keeping labeling literal and consistent with the dashboard's analytical function.
- Decorative quotes and taglines originally included on the landing page were removed to keep the landing page focused and professional rather than marketing-oriented.

Summary

The final design system prioritizes consistency and clarity: a unified color palette applied across all pages, consistent typography sizing between KPI cards and charts, and a simplified navigation structure — resulting in a cohesive, professional visual identity appropriate for an analytical dashboard project.

Link: [Figma](#)

6.4 Dashboard

The following sections present each dashboard page with its full screenshot and a detailed description of the visuals, KPIs, and insights it delivers.

6.4.1 Cover Page

Overview

The landing page serves as the cover/opening page of the project, introducing the dashboard before the viewer proceeds to the data pages. It follows the same visual system as the rest of the dashboard — dark background, warm accent palette, and serif/mono typography — to establish visual consistency from the very first page.

Layout

The page is structured as a single, centered composition with no scrolling, consistent with the dashboard's static, presentation-style format:

- **Title:** "Bitcoin Analysis Project" is displayed as the primary heading, using the serif display typeface at a large scale to anchor the page.
- **Introductory statement:** A short, professional overview statement is placed beneath the title, describing the purpose of the project in plain, factual language.
- **Founders section:** Below the introductory text, the page features a section introducing the four project members, displayed as overlapping circular photo cards arranged at slight alternating angles. Each card links to the corresponding founder's LinkedIn profile.

Visual Treatment

- **Background:** Deep plum-black base with two soft blurred glow gradients (magenta and violet) positioned behind the main content, plus a low-opacity grain overlay for added texture and depth.
- **Color palette:** Gold for accents and data-related elements, magenta and violet for emphasis, cream for primary text, and muted lavender-grey for secondary text — consistent with the palette used across all dashboard pages.
- **Typography:** A serif display typeface for the title, a clean sans-serif for supporting text, and a monospace typeface used sparingly for small data-style labels, maintaining the same type hierarchy used throughout the dashboard.

Purpose

The landing page's role is purely introductory — it establishes the project's identity and visual tone, introduces the team, and transitions the viewer into the dashboard's analytical content, without including any interactive elements, navigation, or functional buttons.



6.4.2 Page 1 — Market Overview

The Market Overview page serves as the opening data page of the dashboard, providing a high-level snapshot of Bitcoin market behavior before the viewer moves into more detailed analysis on later pages. It presents price movement and volatility through a candlestick-style chart, alongside supporting KPI cards covering high-low range, consolidation range, and compression signals — giving a quick read on both the direction of the market and how actively it's fluctuating. The page follows the same visual system as the rest of the dashboard, using the shared color palette, typography, and left-hand navigation to maintain consistency, while focusing specifically on the market's overall pulse before deeper dimensions like buyer/seller pressure or trader activity are introduced on subsequent pages..

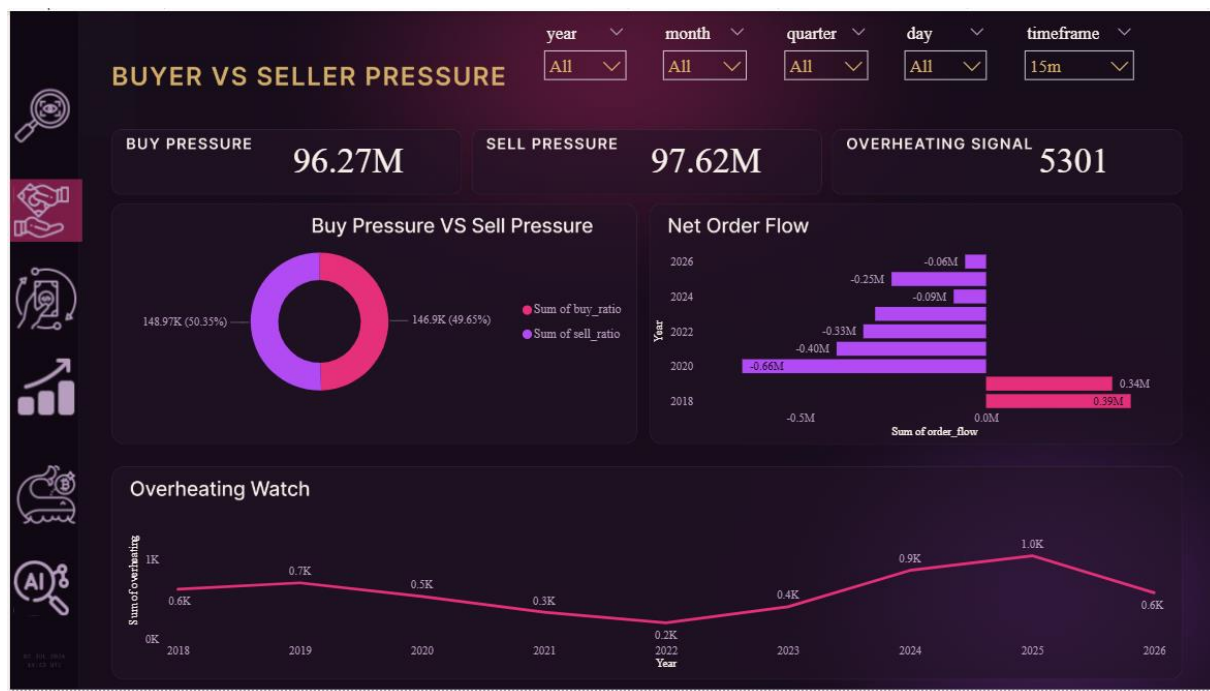
Element	Type
Avg. volume	KPI Card
Avg. Capital Flow	KPI Card
Return_Pct	KPI Card
Avg. Volatility	KPI Card
Price Movement	Line Chart
Volatility Range	Area Chart
Volume Spikes	Clustered Column Chart
Capital Flow	Line Chart



6.4.3 Page 2 — Buyer VS Seller Pressure

The Buyer vs Seller Pressure page examines the ongoing balance between buying and selling activity in the Bitcoin market, using taker buy and sell volume ratios to show which side is currently dominant. A net order flow chart tracks this balance over time, highlighting the moments where buying or selling pressure shifts, while an overheating signal flags periods where rapid price increases coincide with unusually high buy volume, indicating potential short-term market imbalance. Together, these visuals give a clear picture of market sentiment and pressure dynamics, following the same color palette, typography, and layout conventions established across the rest of the dashboard.

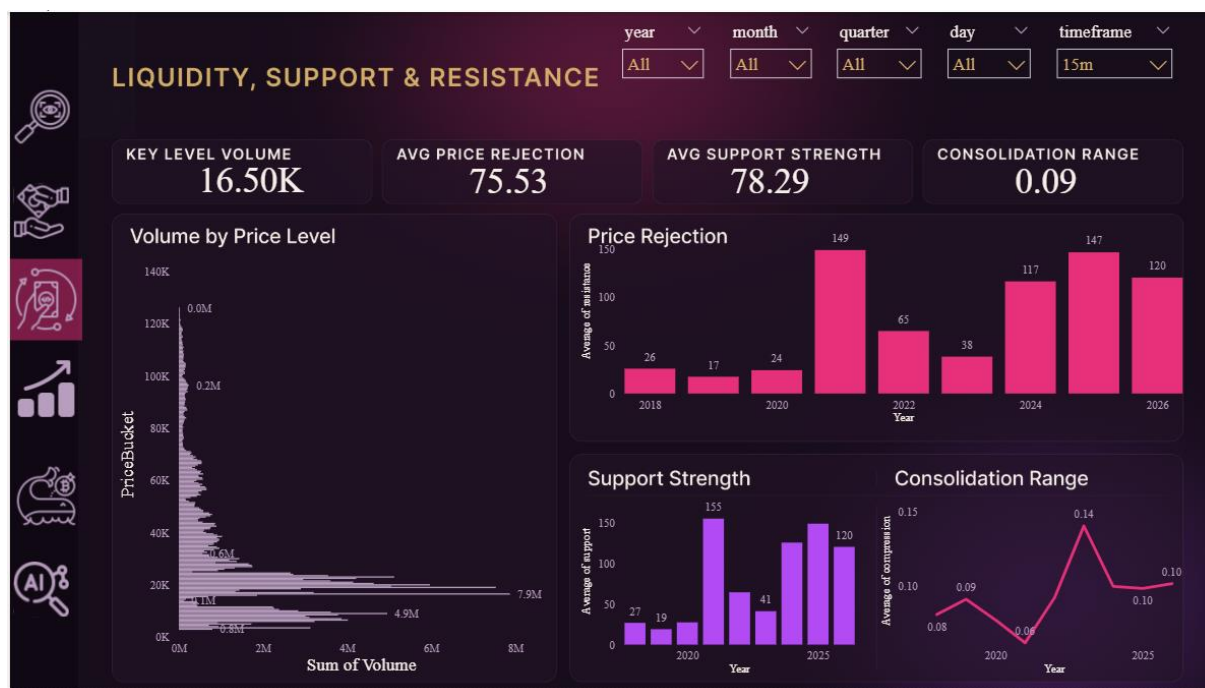
Element	Type
Buy Pressure	KPI Card
Sell Pressure	KPI Card
Overheating Signal	KPI Card
Buy Pressure VS Sell Pressure	Donut Chart
Net Order Flow	Clustered Bar Chart
Overheating Watch	Line Chart



6.4.4 Page 3 — Liquidity, Support & Resistance

The Liquidity, Support & Resistance page focuses on where trading activity concentrates within the market and how price behaves around key levels. A volume-by-price-level chart identifies the price zones with the highest trading interest, highlighting areas of strong liquidity, while price rejection and support strength metrics show where the market has repeatedly failed to break through resistance or has been consistently defended by buyers. A consolidation range indicator rounds out the page, signaling periods where the market is trading sideways rather than trending. As with the rest of the dashboard, the page maintains the same color palette, typography, and layout structure to ensure visual consistency throughout the project.

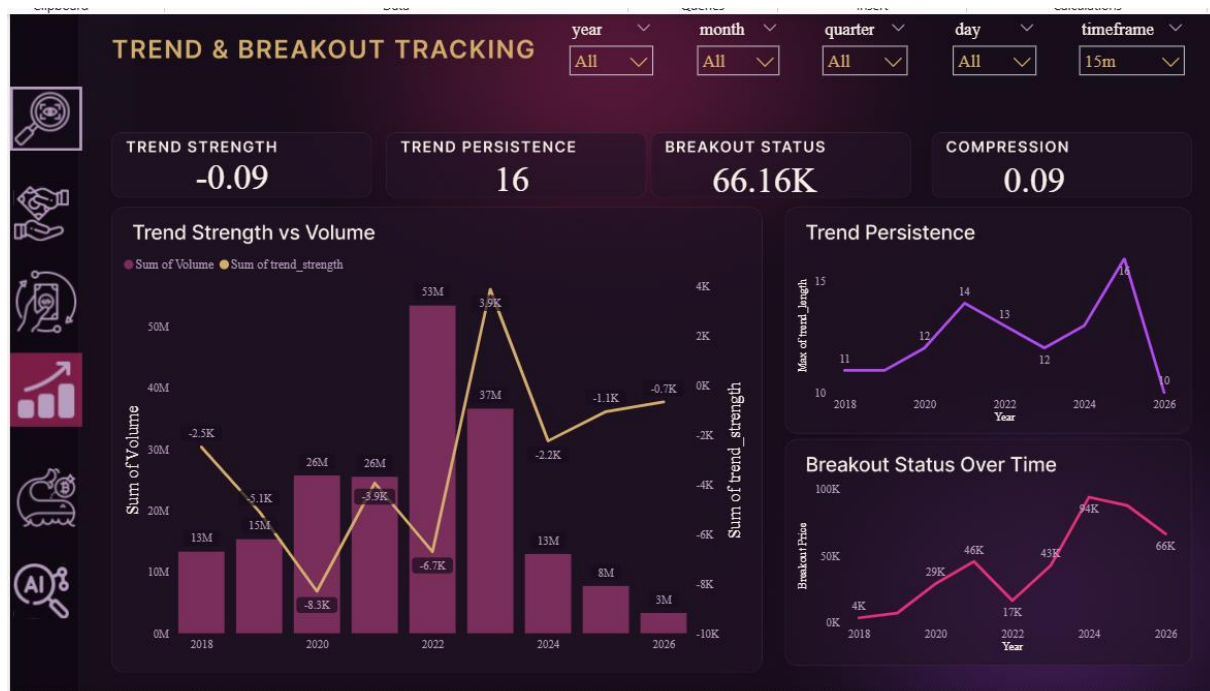
Element	Type
Key Level Volume	KPI Card
Avg Price Rejection	KPI Card
Avg Support Strength	KPI Card
Consolidation Range	KPI Card
Volume By Price Level	Clustered Bar Chart
Price Rejection	Clustered Column Chart
Support Strength	Clustered Column Chart
Consolidation Range	Line Chart



6.4.5 Page 4 — Trend & Breakout Tracking

The Trend & Breakout Tracking page assesses whether current price movements reflect genuine market momentum or short-lived fluctuations. Trend strength is measured relative to trading volume to determine whether a move is backed by real participation, while trend persistence tracks how many consecutive intervals continue in the same direction to gauge the durability of that trend. A breakout versus compression view rounds out the page, distinguishing between periods where price is coiling within a narrowing range and moments where it breaks decisively beyond recent highs or lows. Consistent with the rest of the dashboard, the page follows the same visual system in color, typography, and layout.

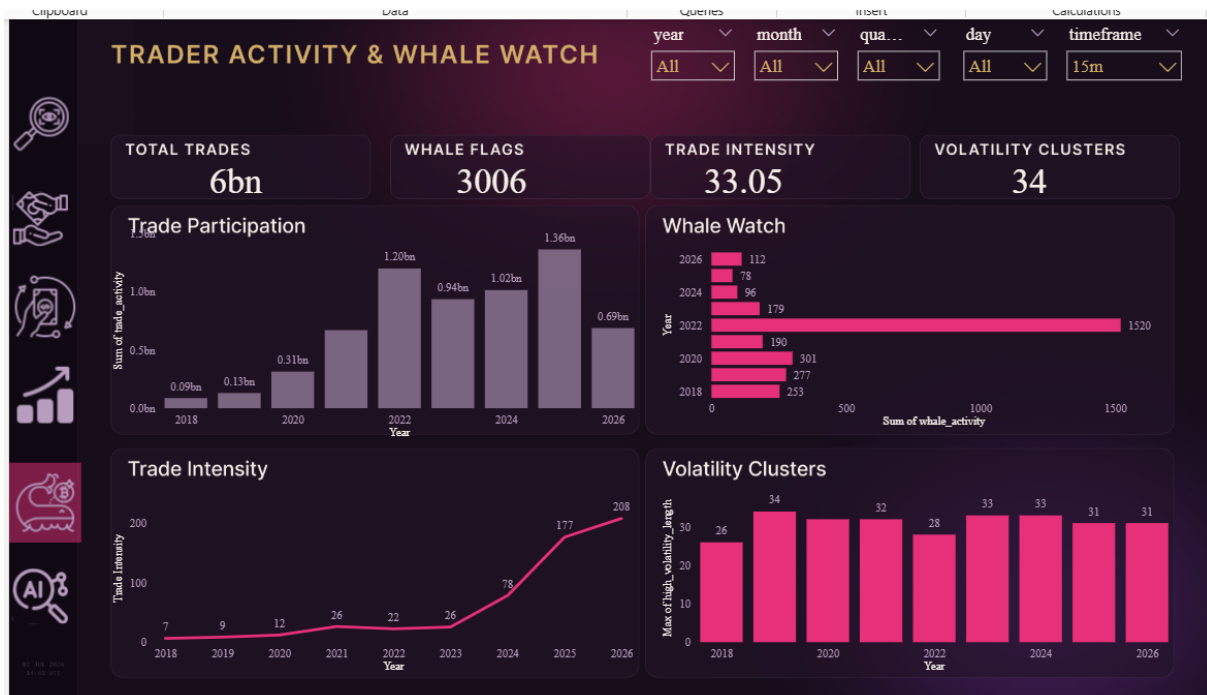
Element	Type
Trend Strength	KPI Card
Trend persistence	KPI Card
Breakout Status	KPI Card
Compression	KPI Card
Trend Strength VS Volume	Line & Clustered Column Chart
Trend Persistence	Line Chart
Breakout Status Over Time	Line Chart



6.4.6 Page 5 — Trader Activity & Whale Watch

The Trader Activity & Whale Watch page focuses on identifying who is actually driving market activity, beyond just price and volume. Trade count and participation metrics establish a baseline for overall market activity, while trade intensity highlights periods of rapid, concentrated trading. Whale flags and a whale watch visualization isolate interval where high trading volume occurs alongside a low number of trades, a pattern typically associated with large, concentrated positions rather than broad market participation. A volatility clusters view complements this by showing how these flagged periods often align with bursts of heightened price movement. As with the rest of the dashboard, the page maintains consistent color palette, typography, and layout conventions.

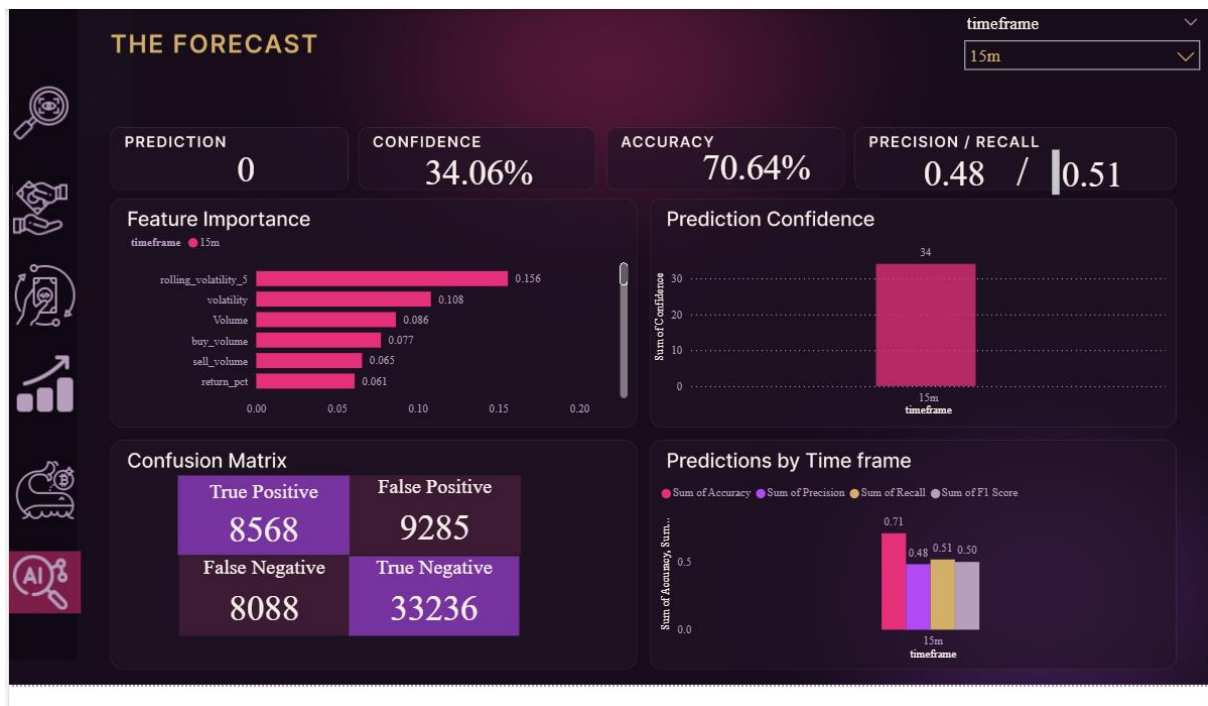
Element	Type
Total Trades	KPI Card
Whale Flags	KPI Card
Trade Intensity	KPI Card
Volatility Clusters	KPI Card
Trade Participation	Clustered Column Chart
Whale Watch	Clustered Bar Chart
Trade Intensity	Line Chart
Volatility Clusters	Stacked Column Chart



6.4.7 Page 6 — The Forecast

The Forecasting page shifts the dashboard's focus from analyzing past market behavior to anticipating future price direction. Building on the patterns established in earlier pages, it presents a projected outlook for Bitcoin price movement, giving viewers a forward-looking perspective alongside the historical analysis covered throughout the rest of the project. The page maintains the same visual identity as the rest of the dashboard, using consistent color palette, typography, and layout to ensure a unified presentation from start to finish.

Element	Type
Prediction	KPI Card
Confidence	KPI Card
Accuracy	KPI Card
Precision/Recall	KPI Card
Feature Importance	Clustered Bar Chart
Prediction Confidence	Clustered Column Chart
Confusion Matrix	KPI Matrix
Predictions By Time Frame	Clustered Column Chart



Link: [Power bi](#)

7

Microsoft Fabric Implementation

7 Microsoft Fabric Implementation

7.1 Workspace Creation

A dedicated Microsoft Fabric workspace was created to manage all project resources in one place. This workspace contains the notebooks, Lakehouse, Data Warehouse, semantic model, reports, and pipeline.

7.2 Lakehouse Creation

A Lakehouse named bitcoin_storage was created to store all project data. The Lakehouse was used as the central storage location for every stage of the ETL pipeline.

7.3 Bronze Layer

A notebook named Bronze was created.

In this stage:

Four Bitcoin historical datasets were loaded from Kaggle.

The datasets represent different market timeframes:

- 15 Minutes
- 1 Hour
- 4 Hours
- 1 Day

The output was stored in the Lakehouse as the Bronze table.

7.4 Silver Layer

A notebook named Silver was created.

This stage focused on data cleaning and feature engineering.

The processed dataset was then saved as the Silver file inside the Lakehouse.

7.5 Gold Layer

A notebook named Gold was developed.

The purpose of this layer was to transform the Silver dataset into a dimensional Data Warehouse using a Star Schema.

Three tables were created:

- Dimension Tables
- Dim_Date
- Dim_Timeframe
- Fact Table
- Fact_Bitcoin_Market

Primary keys and foreign keys were generated to connect the dimensions with the fact table.

The Gold tables were saved inside the Lakehouse .

7.6. Machine Learning Layer

A notebook named ML was created.

This notebook performs Bitcoin market prediction using a Random Forest Classifier.

The process includes:

- Loading Fact and Dimension tables.
- Creating lag features.
- Creating rolling average features.
- Generating target labels based on future price movement.
- Splitting the data into training and testing sets.
- Training a Random Forest model separately for each timeframe.
- Evaluating the model using:
 - Accuracy
 - Precision
 - Recall
 - F1 Score
 - Confusion Matrix
- Generating live Buy / No Buy predictions.
- Calculating feature importance.

The notebook stores the following output tables:

- ML_Predictions
- ML_Metrics
- Feature_Importance
- Confusion_Matrix

These tables are also saved into the Lakehouse.

7.7 Semantic Model

A semantic model was automatically generated from the Data Warehouse.

Relationships between the fact and dimension tables were verified to ensure proper filtering and aggregation in Power BI.

7.8 Power BI Report

A Power BI report was created using the semantic model.

The report contains multiple dashboards for:

- Market Overview
- Buyer & Seller Pressure
- Liquidity, support & resistance
- Trend & breakout tracking
- Trader activity & whale watch
- Forecast
- Interactive slicers allow filtering by timeframe and date.

7.9 Data Pipeline

A Microsoft Fabric Pipeline was created to automate the complete workflow. The pipeline executes the following sequence:

- Bronze Notebook
- Silver Notebook
- Gold Notebook
- Machine Learning Notebook
- Refresh the Semantic Model

This ensures that whenever new Bitcoin data becomes available, the entire pipeline runs automatically, updates the Lakehouse, rebuilds the Data Warehouse, retrains the machine learning model, refreshes the semantic model, and updates the Power BI dashboards without manual intervention.

Link: [Fabric](#)